Sample Resume – Ph.D. Electrical/Computer/International Student

92 Elm St.
Apt # 6
Greenbelt, MD 20770

Farva Ahadi

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in/farvahadi

Education

Ph.D. in Electrical and Computer Engineering, Communication, GPA 3.91 — Expected: 5/2024
University of Maryland, College Park, MD

M.S. in Electrical Engineering, Control Systems, GPA 3.81 — 9/2018
B.S. in Electrical Engineering, GPA 3.78 — 5/2016
Sharif University of Technology, Tehran, Iran

Computer Skills

Platforms: Unix, Linux, DOS, Windows XP/2000/NT, and VAX/VMS
Languages: C/C++, Java, MATLAB, Verilog, Assembly and C for Texas Instruments DSP processors, Assembly and C for embedded systems and Intel x86 Assembly
Software: Network Simulator (**NS2**) GloMoSim, CPLEX, and Qualnet

Experience

Graduate Research Assistant, ECE Department, College Park, MD — 8/2020 - Present

- Conduct research on dynamical behavior of TCP traffic in IP networks, and developed ***award winning*** CDMA Aggregate Perturbation **(CAP)** technology for Distributed Denial of Service (DDoS) Internet attacks (***C/C++, MATLAB, TCL and NS2 code***).
- Lead the design team on a library of signal processing blocks in Verilog. Designed and implemented ***Dataflow/RTL*** and ***gate level*** realization DSP blocks including FIR and IIR filters.
- Implement both the transmitter and receiver of a V22bis modem according to the ITU-T recommendations based on the Texas Instruments TMS320C30 DSPs (***C and TI Assembly Code***).

Control System Designer, MKK Control Systems (founder), Tehran, Iran — 8/2018 - 7/2020

- Designed front-end of an embedded system of an autonomous process controller, which is currently being mass produced and has been installed in more than 100 plants.

Control System Design Chief Engineer, Fan-Niroo Company, Tehran, Iran — 8/2018 - 8/2020

- Designed and implemented a control, emergency shutdown and process visualization system.
- Oversaw extensive hardware design of digital and analog control boards and implementation of programming in C/C++ and X86 Assembly.

Awards and Leadership

- ***First Place Award of Business Plan Competition,*** University of Maryland, 2020, for **MacroPhage Networks** (With Prof. M. Shayman and Dr. M. Alasti).

- ***Received $50,000 University Technology Development Fund (UTDF),*** Maryland Technology Development Corporation (TEDCO), 12/2019. (With Prof. M. Shayman).

- ***President,*** University of Maryland Electrical and Computer Engineering Graduate Student Association (ECEGSA), 2019 -2020.

U.S. Patent and Invention Disclosure

- ***Method for Quantifying Responsiveness of Flow Aggregates to Packet Drops in A Communication Network*** (US pending patent number 123456789).

- ***Using Direct Sequence Spread Spectrum to Determine Responsiveness of a TCP Aggregate to Packet Drops,*** reported to the Office of Technology Commercialization, University of Maryland, 4/2020, Ref. No. IS-1111-000.

Selected Publications

M. Shayman, R. Gahremanpour, R. Skoog, N. Jasinski and **Farva Ahadi,** "*Network Management and Control Mechanisms to Prevent Maliciously Induced Network Stability*," Proc. 8th IEEE/IFIP Network Operations and Management Symposium (NOMS-2021).

Farva Ahadi, K. Gallichio, and M. Shayman, "*Mitigation of Denial of Service Attacks in the Internet*," Proc. 41st IEEE Conference on Decision and Control (CDC-2018).

Research Proposals /Grants

"Routing and Topology Design of Hierarchical Sensor Networks" With Prof. Mark Shayman, ECE Department of the University of Maryland, Submitted to NSF Sensornet program 1/2020.

"CDMA-Based Mitigation of Distributed Denial of Service Attacks" With Prof. Mark Shayman, ECE Department of the University of Maryland, Submitted to NSF NetS program 4/2019.

Professional Activities/Affiliations

- Paper Reviewer, INFOCOM 2020
- Paper Reviewer, International Conference on Communication (ICC) 2019 and 2021
- Member, Scientific Research Society (Sigma Xi)
- Student Member, IEEE

Relevant Graduate Courses

University of Maryland: Random Processes in Communications and Control, Multi-User Communication, Wireless Communication, Digital Communications, Detection and Estimation Theory, Digital Computer Design, CAD of Digital Systems, Advanced Digital System Design

Sharif University of Technology: Switching Systems, Information Theory, Data Communication Networks, Object Oriented Programming, Neural Networks, Fuzzy Systems and Sets, Adaptive Control, Multi Variable Control, Optimal Control, Robust Control, Robotic Manipulators, Nonlinear and Digital Control, Discrete Signal Processing, Operation Research, Abstract Algebra

Additional Information

In the Media: "UM Business Plan Competition Could Launch Next Google," 5/3/2020. Received favorable comments about MacroPhage Networks and the CAP technology. Covered by PR Newswire, CBS MarketWatch, NBC, National Hispanic Corporate Council, and The Gazette.